INCH-POUND

MIL-DTL-6363/5A
30 September 1997
SUPERSEDING
MIL-L-6363/5
13 September 1993

DETAIL SPECIFICATION SHEET

LAMPS, INCANDESCENT, AIRCRAFT SERVICE, SUBMINIATURE UNBASED WIRE TERMINALS, INTEGRAL LIGHTING T-1 BULB

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-6363.

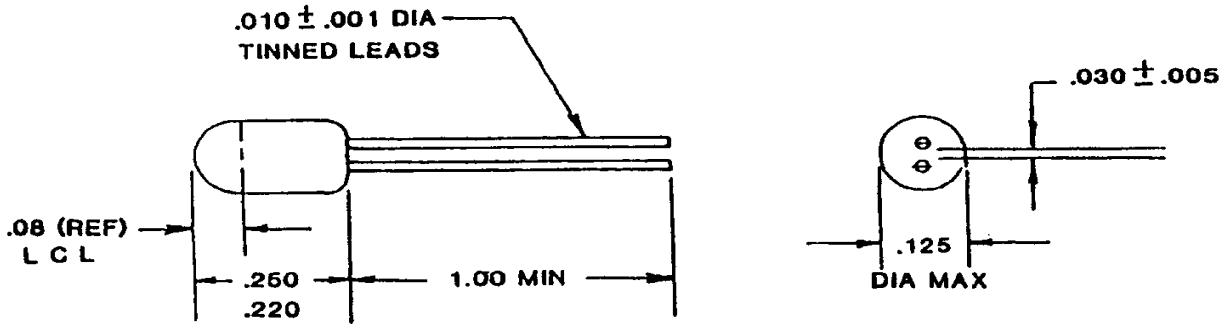


FIGURE 1. Full envelope.

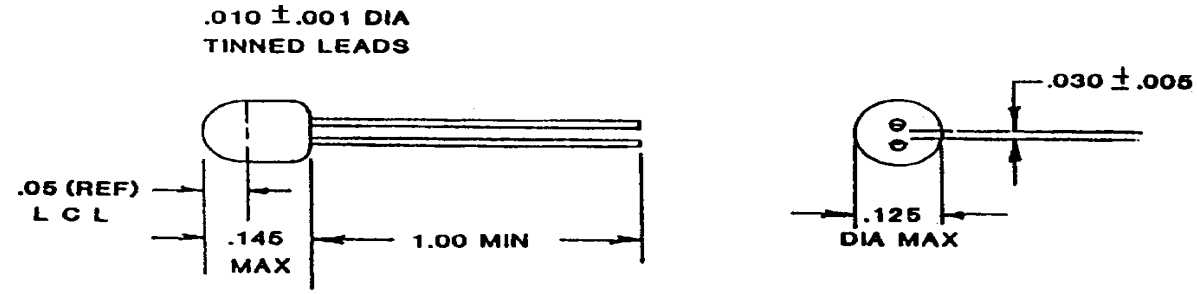


FIGURE 2. Short envelope.

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NOTES:

1. Dimensions are in inches.
2. Intended use - integral lighting for aircraft instruments, indicator lights, and light panel assemblies.
3. For design feature purposes, this specification takes precedence over procurement documents referenced herein.
4. Qualification by similarity - Lamp M6363/5-8 is required to be fully qualified in accordance with table I of MIL-DTL-6363. Lamp M6363/5-10 may be qualified by similarity in accordance with table II of MIL-DTL-6363. All other "AS15" designated lamps may be granted qualification by similarity and extension of qualification from the qualified lamps with no additional testing. Red lamp M6363/5-8R may be qualified in accordance with table II of MIL-DTL-6363. All other red lamps may be granted qualification by similarity and extension of qualification from M6363/5-8R with no additional testing. Refer to MIL-DTD-6363 for conditions and data requirements for qualification.

REQUIREMENTS:

1. Configuration and dimensions: See figures 1 and 2.

2. Finish: Glass globe finish shall be as indicated in table I.

3. Lamp operating characteristics: When operating at rated voltage, the lamp operating characteristics shall be as specified in table I.

4. Where indicated by suffix "AS15" in the part number, lamps shall be aged for a minimum of ten hours at rated voltage and selected for 15 percent tolerance on MSCD as indicated in table I.

5. MSCD at 1,000 hours shall be not less than 80 percent of initial MSCD when measured at rated voltage.

6. Coating adhesion: Applicable to exterior coated colored lamps.

7. Solderability: Applicable.

8. High temperature: Applicable for qualification tests only (85°C/185F°).

9. Thermal shock: Applicable for qualification tests only. Use temperatures for Type I lamps.

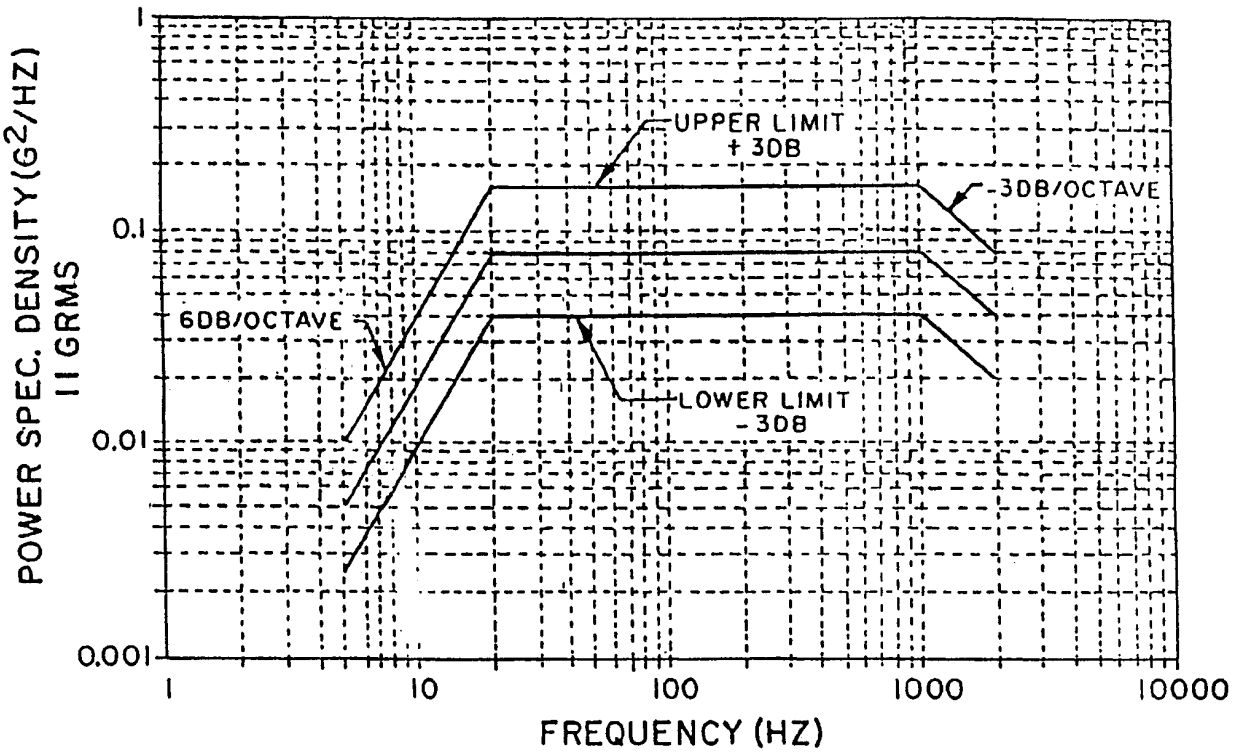
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TABLE I. Part or identifying numbers (PIN) and operating characteristics.

Part or Identifying Number (PIN)	Figure Number	Type MIL-DTL-6363	Bulb Finish	Shock Test Level (G's)	Electrical Ratings			Filament Style	MSCD ±15 % 1/	Average Rated Lab Life at DC
					Volts (Nom)	Amperes (±10%)	Watts (Max)		Candelas	Hours
M6363/5-1 AS15	1	I	Clear	50	5.0	0.060	0.33	C-2R	0.03	30,000
M6363/5-2 AS15	1		Clear			0.060	0.33	C-2R	0.05	20,000
M6363/5-2R	1		Red			0.060	0.33	C-2R	0.05±25%	20,000
M6363/5-3 AS15	1		Clear			0.115	0.63	C-2R	0.15	20,000
M6363/5-4 AS15	1		Clear			0.060	0.33	C-2R	0.15	2,000
M6363/5-5 AS15	2	I	Clear	50	5.0	0.060	0.33	C-2R	0.03	50,000
M6363/5-6 AS15	2		Clear			0.060	0.33	C-2R	0.05	20,000
M6363/5-7 AS15	2		Clear			0.075	0.41	C-2R	0.09	20,000
M6363/5-8 AS15	2		Clear			0.115	0.63	C-2R	0.15	20,000
M6363/5-8R	2		Red			0.115	0.63	C-2R	0.15±25%	20,000
M6363/5-9 AS15	2		Clear			0.060	0.33	C-2R	0.15	2,000
M6363/5-10	2		Clear		28.0	0.020	0.62	CC-2F	0.15±25%	1,000

1/ MSCD for colored lamps applies to clear lamps before coating.

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Freq Hz	Vibration Level PSD (G^2/Hz)	Upper Limit (+3 DB) PSD (G^2/Hz)	Lower Limit (-3 DB) PSD (G^2/Hz)
5	0.005	0.01	0.0025
10	0.02	0.04	0.01
20	0.08	0.16	0.04
1000	0.08	0.16	0.04
2000	0.04	0.08	0.02

FIGURE 3. Random vibration curve for T-1 lamps.

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10. Random vibration: Applicable for qualification tests only. See figure 3 for random vibration curves. For vibration tests, the lamps shall be potted in a clear epoxy block which shall be rigidly mounted to the vibration table.

11. Shock: Applicable for qualification tests only. See table I for applicable “g” level. For shock tests, the lamps shall be potted in a clear epoxy block which shall be rigidly mounted to the shock machine.

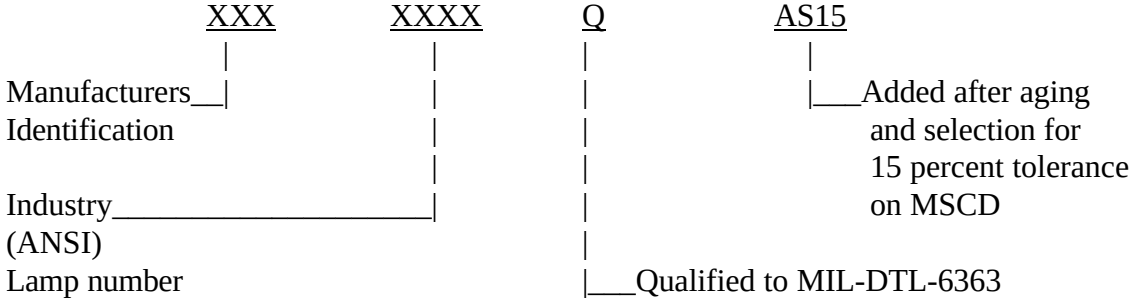
12. Humidity: Applicable. for qualification tests only.

13. Salt spray: Applicable for qualification tests only.

14. Marking: Lamps do not require marking. Unit packages and containers shall be marked as specified in section 5 of MIL-DTL-6363.

Production lamps qualified to MIL-DTL-6363 and listed on QPL-6363 shall be identified on packages and containers in accordance with ANSI-C78.390 and as supplemented herein. ANSI-C78.390 authorized suffix letters shall be supplemented with the letter “Q” (MIL-DTL-6363 QPL item). The letter “Q” shall be incorporated into the method of designation in accordance with the following example:

EXAMPLE



15. Interchangeability: These lamps may be similar to, but are not interchangeable with, other commercial or military lamps due to unique aircraft applications and qualification.

There are no technical changes in this revision.

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review activities:
Navy - MC, SH
Air Force - 11, 82
DLA - GS

Preparing activity
Navy - AS
(Project 6240-1441-05)

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